CANCELLATION: SEPARATEDNESS AND QUASI-SEPARATEDNESS PASS TO THE FIRST FACTOR

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4563-4581, following the statement at 4553-4561.

Allowed standard prerequisites (editorial).

- Fibre products after a morphism stay immersions. lemma-fibre-product-after-map (same chapter, cited at ~4567). X x_Y X -> X x_Z X is an immersion.
- An immersion with closed image is a closed immersion. lemma-immersion-when-closed (card L26391, cited at ~4571). Concludes X -> X x_Y X is a closed immersion.
- Affine characterization of quasi-separated. lemma-characterize-quasi-separated (same chapter, cited at ~4580). Concludes the quasi-separated case from finite affine unions.

Source reference. Original lemma 01KO: sources/stacks-schemes/pinned-original-source.tex, source lines 4342–4342. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01KR: sources/stacks-schemes/pinned-original-source.tex, source lines 4455–4455. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01IQ: sources/stacks-schemes/pinned-original-source.tex, source lines 2066–2066. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of schemes. If $g \circ f$ is separated then so is f. If $g \circ f$ is quasi-separated then so is f.*

Proof. Assume that $g \circ f$ is separated. Consider the factorization $X \to X \times_Y X \to X \times_Z X$ of the diagonal morphism of $g \circ f$. By Lemma 01KR the last morphism is an immersion. By assumption the image of X in $X \times_Z X$ is closed. Hence it is also closed in $X \times_Y X$. Thus we see that $X \to X \times_Y X$ is a closed immersion by Lemma 01IQ.

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Assume that $g \circ f$ is quasi-separated. Let $V \subset Y$ be an affine open which maps into an affine open of Z. Let $U_1, U_2 \subset X$ be affine opens which map into V. Then $U_1 \cap U_2$ is a finite union of affine opens because U_1, U_2 map into a common affine open of Z. Since we may cover Y by affine opens like V we deduce the lemma from Lemma 01KO. $\square$